

India turns to AI for inclusive language integration



In a bid to bridge linguistic gaps, India is utilising AI to capture its 121 languages. Villagers in Karnataka are contributing to a revolutionary chatbot project in Kannada, a language spoken by over 40 million Indians. The government's digital drive, aided by the *Bhashini* translation system, is pioneering open source data sets, breaking barriers in AI accessibility.

For weeks, villagers in Karnataka have actively participated in a project, reading sentences in Kannada to build India's first AI-based chatbot for tuberculosis. Kannada, an official language, plays a pivotal role in this healthcare initiative.

Linguistic diversity poses a challenge for AI, leaving millions without access to information and economic opportunities. Crowdsourcing efforts, including the *Bhashini* translation system, address these gaps by creating open source data sets in local languages for diverse AI applications.

Pushpak Bhattacharyya, from the Computation for Indian Language Technology Lab, acknowledged challenges in collecting data for less common languages due to an oral tradition and limited electronic records. Despite these hurdles, economic value and inclusivity are driving forces behind incorporating diverse languages into AI models.

Safiya Husain, co-founder of Karya, the world's first data co-operative, highlighted the economic potential of crowdsourcing, stating, "We see huge potential for adding economic value with speech data." Collaboration with non-profit organisations ensures workers below the poverty line contribute to data generation, potentially leading to community-centric AI products.

As India pioneers language integration into AI, ethical and inclusive approaches are crucial for scalability and effectiveness. The journey to linguistic inclusion in the AI landscape is challenging, but the potential benefits for millions of Indians are vast and promising.

EndeavourOS releases 'Galileo'

EndeavourOS' latest release called 'Galileo' operates on a rolling release model, offering a continuous flow of updates post a one-time installation.

Microsoft's Azure RTOS to morph into open source Eclipse ThreadX

Azure RTOS, featuring the esteemed ThreadX real-time operating system, will now be known as Eclipse ThreadX under the Eclipse Foundation's stewardship. The move promises to reshape embedded development with a mature, scalable, and safety-certified real-time operating system under a permissive licence.

ThreadX, renowned for its maturity and scalability over its 20-year history, runs on over 12 billion devices globally. The Eclipse Foundation has lauded this move as a solution to the fragmented embedded software market by offering the world's first open source real-time operating system. Eclipse ThreadX will now be open sourced under the MIT licence and governed by the Eclipse Foundation, ensuring a vendor-neutral model.

Certified for functional safety and security, ThreadX holds IEC 61508, IEC 62304, ISO 26262, and EN 50128 conformance, along with EAL4+ Common Criteria security certification—an industry first for open source RTOS.

Mike Milinkovich, executive director of the Eclipse Foundation, labelled the move as an "industry game-changer." William Lamie, president of PX5, applauded Microsoft's contribution in making "one of the best commercial RTOS free for all developers."

Eclipse ThreadX components, including ThreadX, NetX Duo, FileX, GUIX, USBX, LevelX, will be available under the MIT licence from Q1 2024. Microsoft said it is committed to providing the project with the latest safety and security certifications.